

Theorem 1.1: formal statement and verified foundations

Falconer distance-gauge formalization project
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Status. This is an incomplete formalization notebook. The main theorem is an explicit proof obligation; it has not passed Comparator and is not registered on Palomar. The results below record the proved foundation.

The exact target

The uploaded manuscript defines

$$h_\theta(r) = r \exp\left(-(\log(1/r))^\theta\right) \quad (0 < r < 1), \quad h_\theta(0) = 0.$$

Its Theorem 1.1 asserts that, for a compact set $E \subseteq \mathbb{R}^2$,

$$\frac{2}{3} < \theta < 1, \quad \mathcal{H}^{h_\theta}(E) > 0 \implies \mathcal{L}^1(\Delta(E)) > 0.$$

$$\Delta(E) = \{\|x - y\| : x, y \in E\}.$$

The formalization uses the Euclidean norm and Mathlib's arbitrary-gauge Hausdorff measure constructor. For radii at least one, the gauge is extended by the identity; this does not affect its small-radius Hausdorff measure. The covering characterization has been proved:

$$\mathcal{H}^{h_\theta}(E) = \sup_{\rho > 0} \inf_{\substack{E \subseteq \bigcup_{n \geq 0} U_n \\ \text{diam}(U_n) \leq \rho}} \sum_{\substack{n \geq 0 \\ U_n \neq \emptyset}} h_\theta(\text{diam}(U_n)).$$

The independent `Challenge.lean` imports only Mathlib and repeats all custom definitions. The solution does not import Challenge. The theorem statement has no auxiliary Frostman, projection, or energy hypothesis.

The geometric foundation

Proposition 1. *For $0 < r < 1$, the gauge is positive and satisfies $h_\theta(r) \leq r$. Consequently $\mathcal{H}^{h_\theta}(E) \leq \mathcal{H}^1(E)$, and countable sets have zero gauge measure. Positive gauge measure supplies two distinct points of E .*

Proof. The exponential is strictly positive. The logarithm and its real power are nonnegative, so the exponential is at most one. Mathlib's gauge comparison theorem gives the measure inequality. Singletons are null for $\mathcal{H}^1$, hence also for the gauge measure; countable subadditivity gives the countable case. A set containing at most one point is countable. $\square$

The all-pairs distance map is continuous. It therefore maps the compact product $E \times E$ to a compact, Lebesgue-measurable distance set. Two distinct points give a positive distance; that fact alone does not give positive length.

The scalar summability foundation

Proposition 2. *For every $s, c > 0$, one has $\log n \leq cn^s$ for all sufficiently large natural numbers n . For every $\theta, c > 0$,*

$$\sum_{n=0}^{\infty} \exp(-cn^{\theta}) < \infty.$$

Proof. Mathlib proves $\log x = o(x^s)$ as $x \rightarrow \infty$ for $s > 0$. Restricting its eventual norm bound to natural numbers proves the first assertion. Applying that assertion with coefficient $c/2$ gives

$$\log n \leq \frac{c}{2}n^{\theta} \implies \exp(-cn^{\theta}) \leq \exp(-2 \log n) = n^{-2}.$$

The inverse-square series converges. The omitted finite initial segment has no effect on summability, so comparison proves the claim. $\square$

Proposition 3. *The branch-counting exponent and filter-error power obey*

$$3\theta - 2 > 0 \iff \theta > \frac{2}{3}, \quad \theta > \frac{2}{3} \implies -8(2\theta - 1) < -1.$$

Consequently, in the theorem's parameter range, every positive coefficient eventually absorbs the logarithm into $n^{3\theta-2}$, and

$$\sum_{n=0}^{\infty} n^{-8(2\theta-1)} < \infty.$$

Proof. The exponent inequalities are linear arithmetic. Apply the logarithmic domination result with exponent $3\theta - 2$, and apply Mathlib's real-power series criterion to the filter-error exponent. $\square$

These results justify scalar convergence once suitable shell energies and filter errors are supplied. They do not establish the geometric estimates that produce those bounds.

The exact rounded parameters

The gain has the proved exact form $\beta_N = c_{\theta}N^{\theta-1}$, with $c_{\theta} > 0$. The ceiling definitions are retained in the block parameter κ_N and tolerance ε_N . The verified lower bounds imply

$$\varepsilon_N N \geq \frac{(c_{\theta}/1000)^2}{10^6} N^{2\theta-1}, \quad \kappa_N^3 N \geq (c_{\theta}/1000)^3 N^{3\theta-2}.$$

Hence the actual rounded filter errors $(\varepsilon_N N)^{-8}$ are summable, and $\kappa_N^3 N$ eventually absorbs every fixed logarithmic cost when $\theta > 2/3$. The full Lemma 2.2 is now proved: all four budget groups hold above one common threshold. The coarse bound $T + 1 \leq 242N$ makes its derivative constants polynomial; logarithmic domination absorbs them into $2^{\varepsilon_N N/8}$. The remaining count budgets follow from the inverse tolerance bound and the growth of $\kappa_N^3 N$. All three terms in the asserted filter bound are now summable: $(\varepsilon_N N)^{-8}$, $N2^{-\varepsilon_N N}$, and $2^{-\kappa_N N/4}$. The geometric removed-mass estimate remains to be proved.

The verified gauge Frostman construction

Proposition 4. *For $0 < \theta \leq 1$ and a compact set E of positive gauge measure, there are a probability measure μ and a constant $C > 0$ such that*

$$\mu(E) = 1, \quad \text{supp } \mu \subseteq E, \quad \mu(\overline{B}(x, r)) \leq Ch_\theta(r) \quad (0 < r < 1).$$

Proof. Positive Hausdorff gauge measure gives positive gauge content. Finite dyadic trees supply nonnegative weights with uniformly positive total mass and saturated covering bounds. Choosing atoms on the compact set and normalizing gives probability measures carried by E . These measures form a weakly compact family, so a subsequence converges to a probability measure carried by E . Portmanteau transfers open-ball bounds to the limit. Comparing an arbitrary radius with consecutive dyadic scales, and applying the proved gauge doubling estimate, yields the closed-ball bound. The probability bound handles radii bounded away from zero. $\square$

Proposition 5. *The resulting measure has no atoms and has finite critical logarithmic energy for every $\gamma \geq 1$:*

$$J_\gamma(\mu) = \iint \frac{(1 + \log^+(1/\|x - y\|))^\gamma}{\|x - y\|} d\mu(x) d\mu(y) < \infty.$$

Proof. The ball bound tends to zero with the radius, proving atomlessness. The formal kernel is infinite on the diagonal. Decompose each pointwise potential into dyadic annuli; the gauge ball bound controls its n -th annular contribution by a constant times

$$(n + 1)^\gamma e^{-cn^\theta}.$$

Logarithmic domination absorbs the polynomial into half the stretched exponential. The resulting summable majorant uniformly bounds the potential. Integrating against the finite measure proves the energy bound. $\square$

The actual scalar Gaussian kernel is also controlled:

$$K_\gamma(r) = \sum_{n \geq 0} (n + 1)^\gamma 2^n e^{-4^n r^2} \leq C_\gamma \frac{(1 + \log(1/r))^\gamma}{r} \quad (0 < r < 1, \gamma \geq 0).$$

A split at the dyadic index nearest the inverse radius bounds the finite prefix and compares the shifted tail with $K_\gamma(1)$. This kernel now also controls the actual Fourier energy described on the next page.

The verified logarithmic Fourier energy

Proposition 6. *For every planar probability measure μ and $\gamma \geq 0$, a finite positive constant depending only on γ satisfies*

$$\int_{\mathbb{R}^2} |\widehat{\mu}(\xi)|^2 \frac{\log^\gamma(e + \|\xi\|)}{\|\xi\|} d\xi \leq C_\gamma(1 + J_\gamma(\mu)).$$

Proof. Mathlib's evaluated Gaussian Fourier transform and Fubini give the precise duality between Gaussian-weighted squared Fourier modulus and the spatial Gaussian kernel. Tonelli sums the non-negative dyadic Gaussian contributions. The proved scalar kernel bound controls their sum by the critical spatial energy. A dyadic scale nearest each high frequency supplies the desired logarithmic weight. At low frequencies, the probability bound controls the Fourier modulus; the extended inverse norm has a proved finite integral on the planar unit ball. Combining the regions proves Lemma 5.2 with a finite explicit constant and the manuscript's literal Fourier phase. $\square$

Separated probabilities and the prepared geometry

Two compact pieces of positive mass are extracted from the atomless Frostman probability. Their normalized restrictions retain gauge ball bounds and all finite logarithmic energies. The positive similarity used in Proposition 5.6 places the pieces in discs of radius $1/400$ inside the unit square, with centers exactly $1/4$ apart. Every cross distance lies in $[0.24, 0.26]$. These geometric and energy preparation statements are proved; the radial Orlicz density preparation remains to be established.

The actual product-distance pushforward has mass one on the all-pairs distance set. Its absolute continuity therefore gives positive length. Preparation multiplies this length by the positive similarity scale, so positivity transfers back to the original set.

The complete summable reconstruction criterion

Lemma 4.2 is now proved for a finite positive measure α and dominated approximants $\tau_N \leq \alpha$ for $N \geq N_0$. Its exact hypotheses are

$$\sum_{N \geq N_0} (\alpha(\mathbb{R}) - \tau_N(\mathbb{R})) < \infty, \quad \sum_{N \geq N_0} \int_{2^{N-1} \leq |r| \leq 2^{N+1}} |\widehat{\tau}_N(r)|^2 dr < \infty.$$

They imply $\alpha \ll \mathcal{L}^1$. The concrete convolution errors and Fourier bands form convergent L^1 and L^2 series. Telescoping the low-pass convolutions identifies their pairing with the original measure. The final absolute-continuity argument is given on the next page.

The verified reconstruction bridges

Dyadic frequency shells overlap at most three times. The quadratic L^2 bound proves convergence of square-summable shell pieces. Actual smooth low-pass kernels preserve mass; their differences have uniformly bounded L^1 norm and the precise dyadic Fourier support. Convolving removed measures with these differences produces summable L^1 errors whenever the removed masses are summable. Low-pass convolutions converge to the original measure on Schwartz tests.

Proposition 7. *Suppose a finite positive measure α satisfies*

$$\int \varphi d\alpha = \int \varphi f dx + \int (\mathcal{F}^{-1}\varphi)g d\xi \quad \text{for all Schwartz tests } \varphi, \quad f \in L^1, \quad g \in L^2.$$

Then α is absolutely continuous with respect to Lebesgue measure.

Proof. Let $d\sigma = |f| dx$. Cauchy–Schwarz and Plancherel give

$$\left| \int \varphi d\alpha \right| \leq \int |\varphi| d\sigma + \|g\|_2 \|\varphi\|_2.$$

Use smooth compact cutoffs equal to one on a compact subset of an open set U , zero off U , and bounded between zero and one. Their L^2 norm is at most $|U|^{1/2}$. Inner regularity yields

$$\alpha(U) \leq \sigma(U) + \|g\|_2 |U|^{1/2}.$$

For a Lebesgue-null set, outer regularity provides open covers whose Lebesgue and σ masses both tend to zero. The bound forces its α mass to be zero. $\square$

The actual L^2 shell pieces and the telescoped series pairing have been constructed, completing this part of the reconstruction proof.

Verified projection foundations

The orthogonal-projection development constructs jointly Borel nonnegative densities for almost every direction when the weighted Fourier energy is finite. Their fibers are the actual projection densities and satisfy the exact Plancherel identity. A smooth low-pass decomposition bounds squared density above an amplitude threshold by the corresponding Fourier tail. Summing logarithmic amplitude layers proves the literal one-dimensional bound

$$\int_{\mathbb{R}} f(t)^2 \log^\gamma(e + f(t)) dt \leq C_\gamma \int_{\mathbb{R}} |\widehat{\nu}(r)|^2 \log^\gamma(e + |r|) dr, \quad \gamma \geq 1,$$

for the actual density of a probability with square-integrable Fourier transform. No extra logarithmic exponent is introduced. Polar Fourier integration now gives the full averaged Lemma 5.3:

$$\int_{S^1} \int_{\mathbb{R}} F(w, t)^2 \log^\gamma(e + F(w, t)) dt dw \leq C_\gamma E_\gamma(\mu).$$

The jointly Borel density is the actual orthogonal projection density for almost every direction.

For radial projections, the inverse-distance-weighted smooth source has an exact ray-density identity with respect to circle arc length. The polar Jacobian cancels the inverse-distance factor. Extending rays to whole lines and applying the orthogonal estimate gives the smooth Orlicz transfer.

Uniform logarithmic bounds give an actual quantitative cutoff estimate:

$$\sigma_n(A) \leq H\lambda(A) + \frac{K}{\log^p(e + H)}, \quad p > 0.$$

Portmanteau transfers this estimate to a weak probability limit. Outer regularity extends it from open sets to arbitrary sets. Letting the cutoff grow proves absolute continuity of the limit. A uniform moment of order two greater than the desired exponent also gives finite lower Orlicz moments for the limit's actual Radon–Nikodym density. The target-specific radial endpoint and uniform spread preparation are now assembled below.

Verified regular decomposition

Lemma 5.8 is proved for the original measure weights. The half-open unit square has the exact terminal-cell count. The construction discards light terminal cells, assigns bottom-up types using original masses, and discards whole light type fibers. The rounded schedule is exact:

$$\Delta = \lfloor \varepsilon N / 4 \rfloor, \quad w = \lfloor \varepsilon N / 8 \rfloor, \quad k = \lceil N / \Delta \rceil.$$

Actual parent-child counts bound each class drop, yielding ten possible choices per sampled level. The type count and (P2) give

$$\#\{\text{types}\} \leq (48/\varepsilon + 1)10^{8/\varepsilon+1} \leq 2^{\kappa N/4}.$$

The retained Borel carriers are disjoint unions of original positive-mass terminal cells, with the exact estimates

$$\mu(G_t) \geq 2^{-\kappa N/2}, \quad \mu\left([0,1)^2 \setminus \bigcup_t G_t\right) \leq 2 \cdot 2^{-\kappa N/4}.$$

Restriction preserves each retained terminal-cell weight. Every coarser restricted cell mass equals its original-weight fiber sum. The sampled-level comparisons interpolate to every depth through N . Normalization multiplies all masses by one factor, so the actual normalized probability satisfies

$$\rho_t(Q) \leq 2^{\varepsilon N} \rho_t(P), \quad 0 \leq n \leq N, \quad P, Q \text{ depth-}n \text{ cells}, \quad \rho_t(P) > 0.$$

This is the source's regularity condition under its finite-depth convention.

Verified maximal-cell excess bounds

The maximal cell mass is an actual attained maximum over the finite cells of the unit square. For every probability carried there, successive values obey

$$M(0) = 1, \quad M(n+1) \leq M(n) \leq 4M(n+1), \quad M(n) > 0.$$

The upper successor bound comes from the four actual children of a parent cell; the lower one follows because every child lies in its parent. The excess function is defined from these actual masses by

$$M(n) = 2^{-n} 2^{-NA(n)}.$$

The successive ratios lie in $[1, 4]$, so their base-two logarithms lie in $[0, 2]$. This proves the exact Lipschitz constant for A .

For a retained carrier G , its original mass is at least $2^{-\kappa N/2}$. Every dyadic cell fits in a ball of radius exactly its side length about its midpoint, so its original mass is bounded by $Ch_\theta(2^{-n})$ without an extra radius factor. Normalizing gives

$$M(n) \leq C 2^{\kappa N/2} 2^{-n} 2^{-Ng(n)}.$$

The constant budget $\log_2 C \leq \kappa N/2$ yields the lower excess estimate. All four conclusions of Lemma 5.10 are proved for the actual restriction:

$$A(0) = 0, \quad |A(n) - A(m)| \leq \frac{|n - m|}{N}, \quad g(n) - \kappa \leq A(n) \leq \frac{n}{N}.$$

The upper bound follows from the Lipschitz bound and $A(0) = 0$.

Verified radial endpoint and uniform spread

The gauge supplies every logarithmic energy order. Smooth compact probability approximations contract their Fourier energies and stay in a common carrier separated from the pins. The averaged orthogonal estimate and the exact smooth polar identity bound the actual joint radial densities at a stronger Orlicz order. Their joint measures converge weakly; the proved Orlicz limit argument identifies the actual Radon–Nikodym density of the limit. Thus the prepared radial projections are absolutely continuous and have finite moments at every desired order for almost every pin. This target-specific argument uses stronger energies than the general single-order Theorem 5.4.

Write the eighth-order radial moment as

$$Q(x) = \int_{S^1} \Phi_8(F_x(w)) dw, \quad \Phi_8(t) = t \log^8(e + t).$$

Joint measurability makes the good-pin sets with bounded moment Borel. Since these sets exhaust a full-measure set, one has mass greater than one half. Inner regularity gives a compact subset with original mass greater than one quarter. Restrict and normalize both sources to these compact sets. Each restricted radial measure is at most four times its original radial measure; the actual Radon–Nikodym densities satisfy the same domination almost everywhere. The elementary scaling estimate is proved:

$$\Phi_8(4t) \leq 4(1 + \log 4)^8 \Phi_8(t), \quad t \geq 0.$$

A common finite bound therefore holds at **every** retained pin in both directions. The new probabilities retain the small separated discs, actual compact carriers, and all logarithmic energies. Their common gauge constant increases by at most four. This proves the uniform spread requirement of Proposition 5.6(c), with the actual radial densities.

The logarithmic small-set estimate is also proved for actual densities:

$$\int_Z F \leq H|Z| + K(\log H)^{-8}, \quad H > 1.$$

For the manuscript’s bad-direction length budget, choosing its inverse square root gives the exact bound by a dyadic exponential plus an inverse eighth power. The finite-test weighted loss is now proved below; the full construction still needs the assembly of each regular piece’s own test list.

Verified entry depth and finite test schedule

Actual regular-part cell masses now give the source's occupied-descendant count, rather than a postulated counting bound. The entry depth is the last integer depth at which the actual excess is at most the gain:

$$c = \max\{n \leq N : A(n) \leq \beta\}.$$

The exact parameter facts and gauge lower barrier prove all the entry bounds of Lemma 9.2, including the post-entry budget:

$$\lfloor \beta N \rfloor \leq c < N/4, \quad 100q < N - c, \quad A(n) > \beta - 1/N \quad (c \leq n \leq N),$$

$$V([c, N]) \geq 2\beta - 2/N, \quad k_c[0, c] \leq \beta + \kappa.$$

The finite interval minima define the actual trimmed budget. Its inclusion monotonicity and split gains are proved from actual minimum plateaux. The constructed split uses marked block minimizers and gives the length margins that pay the positive costs of the three moves.

Contributing tube and projection tests are concrete finite records with anchors, endpoints and lengths. Their scheduled lists run over the actual long subintervals, are monotone under interval inclusion, and retain each contributing origin. Anchor depths lie in the constructed marks and every length is at most half its origin's length. The mark count and the exact parameter budget give at most N^2 tests per list, proving Lemma 8.3(b,c). The actual chains and finite trees are now constructed below. Their analytic energy recurrences remain to be proved.

Verified uniform nonstationary estimates and smooth bumps

For a smooth amplitude with derivative bounds

$$\|a^{(j)}\|_\infty \leq AM^j \quad (j \leq K), \quad M \geq 1,$$

assume the actual phase derivative has norm at least a positive number on an open neighborhood of the amplitude's closed support, and its higher derivatives through order $k + 1$ have norm at most a common constant. A globally smooth regularized reciprocal agrees with the true inverse there. Exact Leibniz coefficients and factorial bounds give the source's polynomial base:

$$\left| \int e^{i\phi} a \right| \leq |\text{supp } a| A \left(\frac{2(k+1)^2 \max\{M, \Lambda/\lambda\}}{\lambda} \right)^k.$$

The smooth compact real-line form and the smooth periodic form are both proved. For the latter, endpoint terms cancel over an actual period, and its length replaces the support length. For a linear phase, the transported amplitude is exactly the k -th derivative times the k -th power of the constant inverse slope; the sharper base M divided by the slope lower bound allows M below one. No unspecified order-dependent derivative constants are used.

For every finite order K there is a genuine smooth, nonnegative, even probability bump with compact support in $[-1, 1]$ and

$$\int_{\mathbb{R}} |\omega_K^{(k)}(x)| dx \leq (\pi^2/6)^k (k!)^2, \quad 0 \leq k \leq K.$$

Its construction convolves normalized boxes of half-widths $6/(\pi^2 j^2)$, with a smooth probability tail inside the remaining support radius. The fundamental theorem of calculus differentiates each box average into its boundary difference. The L^1 cost is exactly the inverse box width, giving the factorial product. The Basel identity makes the remaining radius positive. This finite-order construction suffices for the later finite-order masks; an infinite-product identity is not claimed.

Verified directional masks and weighted filter loss

The tube count uses actual occupied dyadic cell centers in the literal Euclidean tube. The projection count uses actual pair coincidence under the normalized anchor-cell restriction. Both are measurable and antipodally symmetric. Markov's inequality gives the exact failure length

$$|\text{Bad}_{\text{test}}| \leq 2\pi 2^{-2\varepsilon N}.$$

Convolving the indicator of a closed neighborhood of an arbitrary passing set with the finite-order bump gives an actual smooth mask between zero and one, equal to one on the passing set. Periodic descent gives a function on the circle. A finite partition into occupied anchor cells proves joint measurability in the pin and direction. Its angular derivative bounds are

$$\|b^{(k)}\|_{\infty} \leq \left(\frac{(4T)^3}{\sigma}\right)^k, \quad k \leq 6T, \quad \sigma = \min\{1, 2^{-\ell} R^{\varepsilon}\}.$$

The weighted distance source and its masked versions are literal measures:

$$\alpha = \text{dist}_{\#}(|x - y|^{-1/2}(\mu_1 \times \mu_2)),$$

$$\tau_N = \text{dist}_{\#}(M_N(x, y)|x - y|^{-1/2}(\mu_1 \times \mu_2)).$$

On the prepared carriers, the weight lies between one and 2.1. Finite lists of at most $N^2 + 1$ genuine tests give a Borel pair mask. The proved regular decomposition supplies the actual carrier defects. The Orlicz estimate then proves domination and the exact summable removed-mass bound:

$$\tau_N \leq \alpha, \quad (\alpha - \tau_N)(\mathbb{R}) \leq a_N,$$

$$a_N = 2.1 \left(4R^{-\kappa/4} + 2 \left[3(N+1)R^{-\varepsilon} + 2^8 K(\varepsilon N)^{-8} \right] \right), \quad \sum_N a_N < \infty.$$

The concrete constructor now assembles every retained regular piece's own list into one Borel filter. Neutral padding uses exactly $N^2 + 1$ shared slots and preserves the actual product. The carrier is the literal regular carrier, so the displayed loss bound applies to the actual scheduled distance measure.

Verified actual chains, trees, and scalar binning

The six actual move types have explicit child states. Every move strictly decreases the interval width, so unfolding terminates. Nodes are genuine chains with proved transitions. A multiset version preserves multiplicities when several paths reach the same state. Exact parameter budgets prove

$$\#\text{ordinary moves} \leq 7/\kappa, \quad \#\text{all moves} \leq 7/\kappa + 1/\varepsilon,$$

$$\#\text{tree visits} \leq R^\kappa.$$

The actual chain costs telescope against the constructed interval minima. For every depth n in its final interval, including short final intervals,

$$\sum \text{cost} \leq 2A(n) + 22\kappa - V(\text{root}), \quad \prod \text{multiplier} \leq R^{\sum \text{cost} + 2\kappa}.$$

Mask levels remain below the source's finite derivative-order bounds. The literal entry-piece list has at most $N^2 + 1$ tests, includes the entry tube, and drops precisely that tube when restricted above the entry depth. The numerical energy-closing and final-shell inequalities are proved from the exact parameter facts. Analytic recurrences for the four actual energies are still required before the tree can prove their estimates.

For a finite positive measure on the line, write

$$C_\eta(h, u) = (\eta \times \eta)\{|s - t - u| \leq h\}, \quad h > 0.$$

Actual half-open bins form a disjoint measurable partition. Discrete Cauchy-Schwarz gives the manuscript's first three binning estimates:

$$C_\eta(h, u) \leq 4C_\eta(h, 0), \quad C_\eta(Lh, 0) \leq (2L + 1)C_\eta(h, 0) \quad (L \in \mathbb{N}),$$

$$\iint (1 + |s - t|/h)^{-2} d\eta(s) d\zeta(t) \leq 10\sqrt{C_\eta(h, 0)C_\zeta(h, 0)}.$$

The decaying-kernel argument uses an actual summable integer-shift majorant. Both Fourier comparisons are now proved as well, as described below.

Verified quadratic phase and circular Morse substitution

The quadratic Fourier identity is proved by a genuine complex Gaussian, followed by dominated convergence as the positive damping tends to zero. The complex square-root branch is evaluated exactly. The purely imaginary Taylor remainder has the sharp factorial bound, with no exponential loss. Every required Fourier moment is an actual integrable Schwartz derivative. Two adjacent derivative norms give an integrable Cauchy majorant of total mass one half in Mathlib's normalized frequency variable.

For Schwartz amplitudes, and therefore for smooth compact amplitudes, the source's exact expansion and error are proved in both signs:

$$\int e^{i\Lambda s^2/2} g(s) ds = \sqrt{2\pi/\Lambda} e^{i\pi/4} \sum_{j < T} \frac{(i/(2\Lambda))^j}{j!} g^{(2j)}(0) + E,$$

$$|E| \leq \sqrt{2/\Lambda} \frac{(2\Lambda)^{-T}}{T!} (\|g^{(2T)}\|_1 + \|g^{(2T+2)}\|_1), \quad \Lambda > 0.$$

This checkpoint claims the smooth-amplitude version used by the constructed masks; the manuscript's full finite-regularity version is not claimed. The literal circular coordinate and its Jacobian satisfy

$$\vartheta(s) = 2 \arcsin(s/2), \quad \vartheta'(s) = (1 - s^2/4)^{-1/2}, \quad \cos \vartheta(s) = 1 - s^2/2.$$

If a smooth cutoff vanishes outside the angular interval of radius $\pi/3$, the actual transformed product with the Jacobian is smooth and supported in $[-1, 1]$. An ordinary integral substitution identifies the localized cosine integral with its quadratic integral. Its finite expansion follows with the actual derivatives of this compact amplitude. Universal real coefficients are defined by finite Leibniz and Faà di Bruno sums of the literal coordinate derivatives. The positive and conjugate operators have proved exact scaled derivative identities, constant term one, and degree at most $2j$. Actual weighted coefficient norms and uniform localized amplitude derivative bounds are now proved, as described below.

Concrete directional averages and assembled filters

The prepared carrier directions lie in one compact closed arc of angular length at most $1/20$. The proof controls the actual relative complex displacement, so it also handles arcs crossing the principal-angle branch.

Nine actual half-open dyadic cells cover each closed disc of dyadic radius, including boundary points. The regular measure's cell bounds apply to the literal normalized anchor probabilities. Circle sine and cosine band estimates, Tonelli, and finite dyadic shells prove Lemma 6.5 exactly:

$$\bar{N}_P \leq R^{\text{ht}[g,p]+4\varepsilon}, \quad \bar{\Pi}_P \leq 2^{-(u-p)} R^{\text{ht}[p,u]+4\varepsilon}.$$

The parameter facts absorb the actual intermediate factors $150(N+4)$ and $75(N+2)$. Passing directions gain the additional factor $R^{2\varepsilon}$. Literal angular widening is proved even at closed-neighborhood boundaries; it does not require the passing set itself to be closed. Each mask derivative vanishes where the next narrower level fails.

The disjoint regular-piece carriers provide a finite Borel pin partition. Each piece contributes its own actual scheduled tests. Neutral padding preserves their product, and the assembled symbol is smooth in the angular variable at every pin. Its derivative scale is bounded by

$$(N^2 + 1)(4T)^3 \max(1, 2^{L-\varepsilon N}),$$

where L is the true maximum length in that piece's finite test list. This supplies the concrete filter data for the proved removed-mass estimate.

Fourier binning and the actual masked Fourier energy

For the literal angular Fourier transform of a finite positive measure, an actual two-box identity gives the remaining binning inequalities:

$$C_\eta(h, 0) \leq 2h \int_{|r| \leq h^{-1}} |\widehat{\eta}(r)|^2 dr,$$

$$\int_{|r| \leq h^{-1}} |\widehat{\eta}(r)|^2 dr \leq 160h^{-1} C_\eta(h, 0).$$

This completes all four parts of Lemma 7.3 without a Fourier identity assumed as a hypothesis. The diagonal remains in the actual collision measure.

The Fourier energy now consists of literal integrals, with the manuscript's actual smooth frequency cutoff Ψ . Its derivative bound is $14^k(k!)^2$ at the selected finite order; its support and plateau give the required dyadic window. For bounded measurable masks the actual U , σ and F satisfy

$$F_{X,Y}(v) \leq 2600 4^v \rho_1(X) \rho_2(Y), \quad F_{X,Y}(v) \leq 2600 4^{v-a} R^{-2A(a)}$$

in the one-piece cell case. A reached mask is a direction-only factor on an actual descendant cell, so removing it can only increase this actual energy. These are the starting bounds; the four nontrivial energy estimates remain.

Uniform localized derivatives and exact inverse operators

The genuine holomorphic Jacobian on complex discs of radius one half obeys Cauchy's estimates. Restriction to the real axis commutes with every iterated derivative, giving, uniformly on $|s| \leq 1$,

$$|J^{(n)}(s)| \leq 2n!2^n, \quad |\vartheta^{(m)}(s)| \leq 2^m(m-1)! \quad (m \geq 1).$$

The cutoff χ_0 is the actual interval indicator convolved with the finite-order bump of radius $\pi/12$. It equals one on the interval of radius $\pi/6$, is supported in the interval of radius $\pi/3$, and obeys $|\chi_0^{(k)}| \leq (2\pi)^k(k!)^2$ through the selected order. Finite Faà di Bruno partitions give the actual localized amplitude g the bounds

$$|g^{(n)}(s)| \leq 2A(n!)^2(56M)^n, \quad \|g^{(n)}\|_1 \leq 4A(n!)^2(56M)^n.$$

The localized expansion has an explicit uniform remainder in universal operator form. The literal coefficient sums satisfy

$$\|P_j\|_M \leq (100jM^2)^j \quad (j \geq 1).$$

The proof counts actual cyclic partition weights, whose sum is exactly $n!$. It does not assume the source's individual coefficient estimate. Literal polynomials define the inverse recursion and prove

$$Q_0 = 1, \quad Q_j = - \sum_{i=1}^j P_i Q_{j-i}, \quad \deg Q_j \leq 2j,$$

$$\sum_{i=0}^j P_i Q_{j-i} = 0 \quad (j \geq 1).$$

Their polynomial multiplication agrees with differential composition on smooth amplitudes, including the conjugate operators. The numerical inverse norms and full periodic expansion for smooth amplitudes are now proved.

Numerical inverse norms and finite inversion

The weighted norm is the actual finite sum of coefficient magnitudes. Its triangle and multiplication inequalities follow from polynomial coefficient expansion. The literal recursion proves the stronger uniform estimate

$$\|Q_j\|_M \leq (200NM^2)^j \quad (j \leq N), \quad \|Q_j\|_M \leq (800jM^2)^j.$$

Applying a polynomial of degree at most d to an actual amplitude consumes d input derivatives and multiplies its amplitude bound by its weighted norm. This is proved at finite differentiability order, also for conjugate operators.

Let C_m be the degree- m coefficient of the two series truncated at T . Actual polynomial convolution gives

$$C_0 = 1, \quad C_m = 0 \quad (0 < m < T), \quad \|C_m\|_M \leq (m+1)(800mM^2)^m.$$

The finite product of the actual differential series equals the input amplitude plus precisely the terms of total order at least T . For a smooth (A, M) -regular amplitude through order $4T$ and $0 \leq q \leq 1/2$, its tail satisfies

$$\|\text{Tail}\| \leq 2A(2T+1)q^T, \quad 800(2T)M^2|z| \leq q.$$

No cancellation or tail estimate is supplied as an analytic hypothesis.

Full periodic circular expansion for smooth amplitudes

The actual periodized cutoff is locally a single translate. Near and opposite cutoffs have disjoint supports. Their complement is smooth, periodic and supported where the sine of the phase angle has magnitude at least $1/2$. The exact circle integral splits into two localized Morse integrals and this actual away integral.

For a smooth periodic amplitude G that is (A, M) -regular through order $2T + 2$, $M \geq 1$ and $\Lambda \geq 1$, the proved expansion is

$$\int_u^{u+2\pi} e^{-i\Lambda \cos(t-\varphi)} G(t) dt = \sqrt{2\pi/\Lambda} \left[e^{-i(\Lambda-\pi/4)} \sum_{j < T} \Lambda^{-j} P_j G(\varphi) + e^{i(\Lambda-\pi/4)} \sum_{j < T} \Lambda^{-j} \bar{P}_j G(\varphi + \pi) \right] + E,$$

$$|E| \leq A \sqrt{2\pi/\Lambda} B (B/\Lambda)^T, \quad B = 10^8 (T+1)^4 M^2.$$

The proof uses $T + 1$ integrations by parts for the away contribution. This retains the square-root normalization and fits the available derivative budget. The manuscript's finite-regularity version is not claimed here; the actual constructed masks are smooth at every order.

The prepared direction arc has a literal convolution cutoff ψ_Γ . It equals one on Γ enlarged by 0.1 , vanishes outside Γ enlarged by 0.2 , and vanishes throughout the opposite arc's neighborhood of radius $\pi/3$. Its angular derivatives obey $33^k (k!)^2$ through the selected finite order.

Actual weighted distance energy and the Fourier bridge

The passing distance measure is the pushforward of the actual restricted product weighted by inverse square-root distance and the passing indicator. Positive separation proves that it is finite. Its normalized collision energy D satisfies both exact trivial bounds (7.2); literal passing sets give monotonicity under level increase, list removal and width enlargement. The actual Fourier transform equals the weighted spatial double integral by measure pushforward and Fubini.

For finite positive measures $\eta \leq \beta \leq \zeta$, normalization $m \geq 0$ and $0 \leq s \leq t$, the genuine low/high bridge is proved:

$$D_\eta(t) \leq 320D_\zeta(s) + \frac{320}{m} \sum_{s < v \leq t} \int_{2^{v-1} \leq |r| \leq 2^v} |\widehat{\beta}(r)|^2 dr.$$

The proof allows real s and includes a partial first dyadic shell. Its stronger high-frequency coefficient is two; the displayed source constant 320 follows. For the actual ordered scheduled lists, the literal masks give passing $\leq$ masked $\leq$ next-level passing measures. This instantiates the bridge at $s = t - \varepsilon N$ and proves Lemma 7.4 with its source constant. The four nontrivial energy estimates are the next analytic assembly.

Concrete symbols and spatial separation

The symbol class is the actual finite product of normalized angular mask derivatives, multiplied by a Borel spatial weight bounded by one. Its order is the sum of the derivative orders. Measurability, the bound by one, and the next-level passing implication follow from these actual factors.

To differentiate symbols of initial order at most $2T$ up to another $6T$ times, the construction chooses the finite bump order $K = 8T$. The existing $6T$ bounds remain valid. Each true derivative has a finite Leibniz expansion into symbols of increased order, with total coefficient magnitude at most

$$M_L^j, \quad M_L = 128T^2|L| \max(1, 2^{\max \text{length}(L) - \varepsilon N}).$$

Dropping reached factors contracts the actual Fourier energy. The state energies in Definition 8.4 are genuine suprema and finite cell maxima of these actual energies; their trivial bounds follow from the actual cell masses.

For balls of radius ρ with centers x_0, y_0 separated by $D \geq 50\rho$, the genuine inverse-distance power has a uniformly convergent separated polynomial series

$$|x - y|^{-j} = \sum_{n=0}^{\infty} \gamma_n p_n(x) q_n(y), \quad |p_n(x)|, |q_n(y)| \leq 1, \quad \sum_n |\gamma_n| \leq (\sqrt{2}/D)^j.$$

Absolute convergence and uniform convergence on the product of the closed balls are proved. This is stronger than the coefficient constant in Lemma 3.9.

The short terminal-state bounds are also actual energy estimates. For a true move chain whose final interval has length at most $100\kappa N$, its actual multiplier product pays the final energy at every mask level:

$$\text{Mult}(\text{chain}) \mathcal{E}_\rho(\text{final}) \leq 2^{N(-V(\text{root}) + 225\kappa)}.$$

This follows from the actual excess bounds and the proved chain cost; no terminal-energy bound is assumed.

One-sided inversion for the actual masks

The prepared cutoff equals one near the true pair directions and vanishes near their opposite directions. Thus the opposite stationary series vanishes term by term. The finite P/Q convolution leaves exactly the high-order tail. The correct complex normalization is

$$c_1 = (\sqrt{2\pi})^{-1} e^{-i\pi/4}.$$

For the actual pair of scheduled symbols, the proved pointwise inversion is

$$\left| d^{-1/2} e^{-ird} B(\varphi) - c_1 \sqrt{r} \sum_{j < T} (rd)^{-j} \int_u^{u+2\pi} e^{-ird \cos(t-\varphi)} \psi_\Gamma(t) Q_j B(t) dt \right| \leq 2^{-200N}.$$

Here $d > 0$, $r > 0$, the true direction lies in Γ , and the source frequency, list-length, cardinality and terminal-distance conditions hold. The actual symbol derivative scale satisfies the needed frequency and terminal budgets, including an empty list. ParameterFacts proves the geometric ratio bound

$$\varpi \leq 2^{-7\epsilon N/8}, \quad \varpi^T \leq 2^{-210N}.$$

The proof retains the square-root normalization in the stationary remainder. No pointwise inversion result is supplied as a hypothesis.

Bilinear Cauchy–Schwarz and countable expansions

For the actual Fourier amplitudes U_X, U_Y , a measurable circle weight bounded by one, and $r > 0$, define the literal integral

$$Z(r) = \sqrt{r} \int \psi(w) U_X(r, w) \overline{U_Y(r, w)} dw.$$

The actual circular Cauchy–Schwarz inequality, including zero cell masses, gives

$$|Z(r)|^2 \leq \rho(X)\rho(Y) \sigma_X(r)\sigma_Y(r)/r.$$

The genuine smooth frequency window equals one on $[2^v/2, 2 \cdot 2^v]$. Integration of this inequality is bounded by $2\rho(X)\rho(Y)F_{X,Y}(v)$. For actual measurable coefficients $|c_n(r)| \leq a_n$ with nonnegative summable a_n , the countable extension proves convergence and the integrated estimate

$$\int_{2^{v-1}}^{2^{v+1}} \left| \sum_n c_n(r) Z_n(r) \right|^2 dr \leq 2\rho(X)\rho(Y) \left(\sum_n a_n \right) \left(\sum_n a_n F_n(v) \right).$$

Dominated convergence justifies the genuine countable series and integral interchange. These are completed analytic tools for Step 3 of Estimate 7.5; the full spatial and operator assembly is recorded below.

Equal arcs, genuine links, and Schur's bound

For $0 < \ell \leq 1$, $M = \lceil 2\pi/\ell \rceil$ equal periodic cells have actual width $w = 2\pi/M$ satisfying $\ell/2 \leq w \leq \ell$ and $M \leq 8/\ell$. The literal source cutoffs are

$$\chi_k = 1_{A_k} * \omega_{w/4}, \quad \sum_k \chi_k = 1, \quad |\chi_k^{(j)}| \leq (576T^2/\ell)^j \quad (j \leq 6T).$$

Their actual support lies within chord distance ℓ of the arc center. The quarter turn is the genuine planar right-angle isometry. True dyadic cell centers and actual points satisfy the necessary phase replacement bounds. For the manuscript's

$$\ell = \min(1, 2^{-p}/2^{-a} \cdot 2^E), \quad \tau = 2^{-p}2^{3E/2}, \quad 2^{E/2} \geq 12,$$

an unlinked pair of cell pairs has either a radial phase or one of the two angular phase derivatives of magnitude greater than $\tau/2$.

For any actual finite symmetric linking graph with row degree at most D , Schur's argument proves

$$\left\| \sum_{\alpha} z_{\alpha} \right\|^2 \leq D \sum_{\alpha} \|z_{\alpha}\|^2 + \sum_{\alpha \not\sim \beta} |\langle z_{\alpha}, z_{\beta} \rangle|.$$

All unlinked inner products are retained explicitly. The following sections prove the actual few-links count and individual cancellation bounds; their joint Fourier-energy assembly remains.

Full spatial and operator assembly of the inverse series

The actual polynomial differential operators are expanded by the finite Leibniz rule. Each branch is a real scheduled symbol multiplied by an actual bounded spatial polynomial. The inverse-distance series converges uniformly on the separated carrier discs, with

$$\sum_n |\gamma_{j,n}| \leq (\sqrt{2}/D)^j, \quad D = \|x_0 - y_0\|.$$

The printed Step 0 bound $2/D \leq 6 \cdot 2^a$ fails at the prepared root $a = 0$, $D = 1/4$. Our proved sharper coefficient bound repairs this step:

$$2/D = 8 > 6, \quad \sqrt{2}/D = 4\sqrt{2} < 6.$$

Both the prepared root and separated cells therefore have the needed scale $6 \cdot 2^a$. Actual Fubini and summability prove that the full inverse-circle integral equals a genuine countable Fourier series. The true branch symbols for an initial mask product have order at most $2T$. On the actual frequency window, nonnegative summable majorants satisfy

$$|c_n(r)| \leq a_n, \quad \sum_n a_n \leq \sum_{j < T} q^j \leq 11/10, \quad q \leq 2^{-7\epsilon N/8}.$$

This completes Step 2 of Estimate 7.5. Its final distance-transform error and frequency-energy assembly remain.

Complete weighted linearization: Estimate 7.7

For the actual separated a -cells and occupied p -cell descendants, with $2p > a + t$, the exact norm Taylor identity gives error at most $2^{-t}/1000$. Four-point distance collisions imply the doubled-width projection collision. The actual passing projection test gives a one-cell raw collision bound of $4B$ times the squared product of cell masses, where

$$B = 2^{-(t-p)} 2^{N(k_p[p,t]+6\varepsilon)}.$$

A witness in a retained cell pair implies that every point in the cell pair passes the shorter lists at the next width. This uses genuine direction motion and the proved width gap. Exact measure sums over retained cells allow grouping by center distances. The true nine-neighbor graph and scalar binning bounds give factors 108 and 9.

The singular weight $d^{-1/2}$ contributes the exact ratio 22. Hence

$$22 \cdot 108 \cdot 9 = 21384 \leq 2^{\varepsilon N},$$

and the actual weighted recurrence is

$$D_{X,Y}^{I,i}(t) \leq 2^{N(k_p[p,t]+7\varepsilon)} D_{X,Y}^{J,i+1}(p).$$

Here $J \subseteq I$, its tests are anchored at depths at most p , and $a + \text{length} \leq p$. The manuscript's stronger length bound $(p - a)/2$ satisfies this condition. No collision or energy estimate is assumed.

Actual nonzero-symbol links and the graph degree

A nonzero built symbol automatically lies on its occupied anchor carrier. Its mask derivative factors provide a genuine passing witness at the next width. Combined with the actual smoothed arc, this selects the contributing fine-cell columns.

Within each g -ancestor group, all linked first-cell centers lie in the passing witness's tube. The exact count gives at most $2^{N(k_p[g,p]+6\varepsilon)}$ centers per group. There are at most $2^{2\varepsilon N}$ groups. For a fixed first cell, the second-cell linking coordinates give $81 \cdot 2^{3\varepsilon N}$ centers. This square bound is coarser than the printed constant 16; the same final exponent holds because $81 \leq 2^{\varepsilon N}$:

$$\deg(\alpha) \leq 2^{2\varepsilon N} 2^{N(k_p[g,p]+6\varepsilon)} 81 2^{3\varepsilon N} \leq 2^{N(k_p[g,p]+12\varepsilon)}.$$

The integrated Schur estimate retains cancellation after each unlinked integral:

$$\int \left\| \sum_{\alpha} Z_{\alpha} \right\|^2 \leq n \sum_{\alpha} \int \|Z_{\alpha}\|^2 + \sum_{\alpha \not\sim \beta} \left| \int \langle Z_{\alpha}, Z_{\beta} \rangle \right|.$$

The actual parent Fourier amplitude is the finite sum over occupied descendants; omitted zero-mass cells contribute exactly zero.

Actual angular and radial cancellation

The true arc support within the period centered on that arc has Lebesgue length at most 2ℓ . This follows from its chord bound $3\ell/4$ and the sine Taylor inequality. Periodic integration by parts uses this support mass, preserving the small-arc factor when summing over arcs.

For the actual amplitude $\chi b(x, \cdot)b(x', \cdot)$, the derivative scale is

$$M_a \leq 1024T^2(N^2 + 1)2^{v-p+\varepsilon N}.$$

An angular linking failure gives an open neighborhood of the true support where the phase derivative is at least $r\tau/3$. Actual cell geometry, $r \geq 2^{v-2}$ and `ParameterFacts` yield

$$\left| \int e^{-irw \cdot (x-x')} \chi(w) b(x, w) b(x', w) dw \right| \leq 2\ell 2^{-90N}.$$

The literal radial amplitude is $h(r) = 2^{-v}\Psi(r/2^v)r^2$, supported in $[2^{v-2}, 2^{v+2}]$. Its actual derivatives satisfy

$$|h^{(j)}(r)| \leq 16 \cdot 2^v (512T^2/2^v)^j \quad (j \leq 6T).$$

For a radial linking failure, the actual linear-phase integral is bounded by 2^{-300N} . The full joint integral, arc sum and normalized F recurrence still need assembly to complete Estimate 7.6.

Actual terminal-tree budgets and the near-cell space split

All terminal visits retain their actual multiplicity. The short-leaf estimate and tree-size budget give

$$\mathcal{T}_{\text{terminal}} \leq 2^{N(-V(\text{root})+226\kappa)}.$$

The actual sum of weighted 2^{-25N} node errors is at most 2^{-20N} . Together they satisfy the bound with 227κ . These results estimate genuine state energies and visits; the analytic recurrence controlling the initial state remains to be assembled.

For the near part of the full space split, true dyadic lattice boxes contain exactly 81 centers within the needed coordinate range. The actual pair-cell near relation is symmetric, with row and column degrees at most

$$81^2 = 6561.$$

The complete far part of Estimate 7.8 still needs the full circular expansion, its two sign combinations and the scalar binning kernel.

Remaining analytic work and verification

Estimate 7.7 is complete. Estimates 7.5, 7.6 and 7.8 need their remaining analytic assembly, followed by the actual induction and frequency shell bound. The main target now uses the proved preparation and weighted distance carrier; its sole hole is absolute continuity of the actual weighted source. Comparator must reject this hole. A successful build is not proof verification.

The public project includes a separate sandboxed Comparator job and a manual full Palomar preflight. Registration requires a completed proof, verification of the exact public commit, truthful metadata, and the registry review.

Public project: [CoolRmal/falconer-theta-gauge](#)